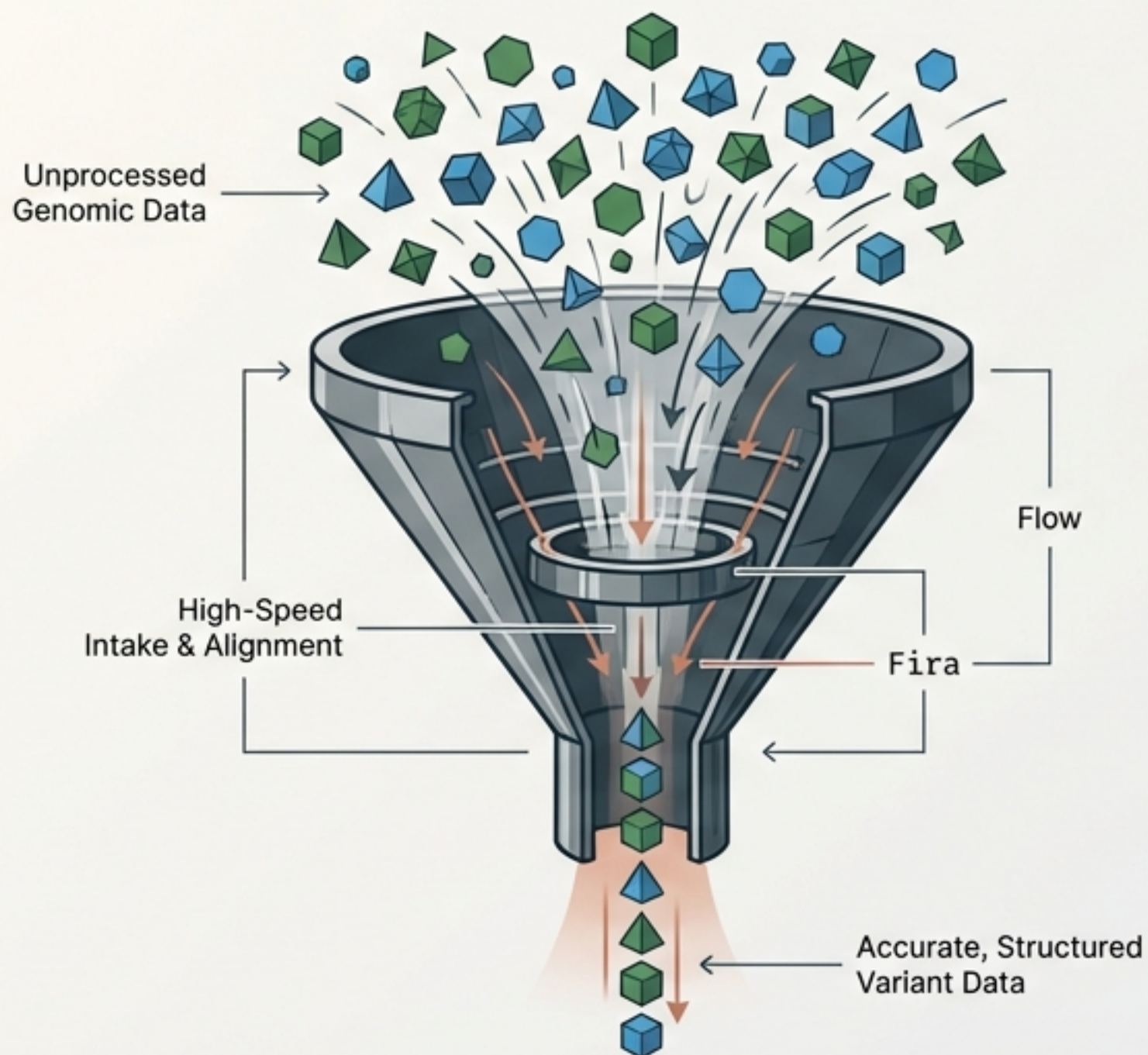


gbcms

The High-Performance Engine for Genetic Variant Counting.

Sifting through the noise of genomic data with blistering speed and ironclad accuracy.

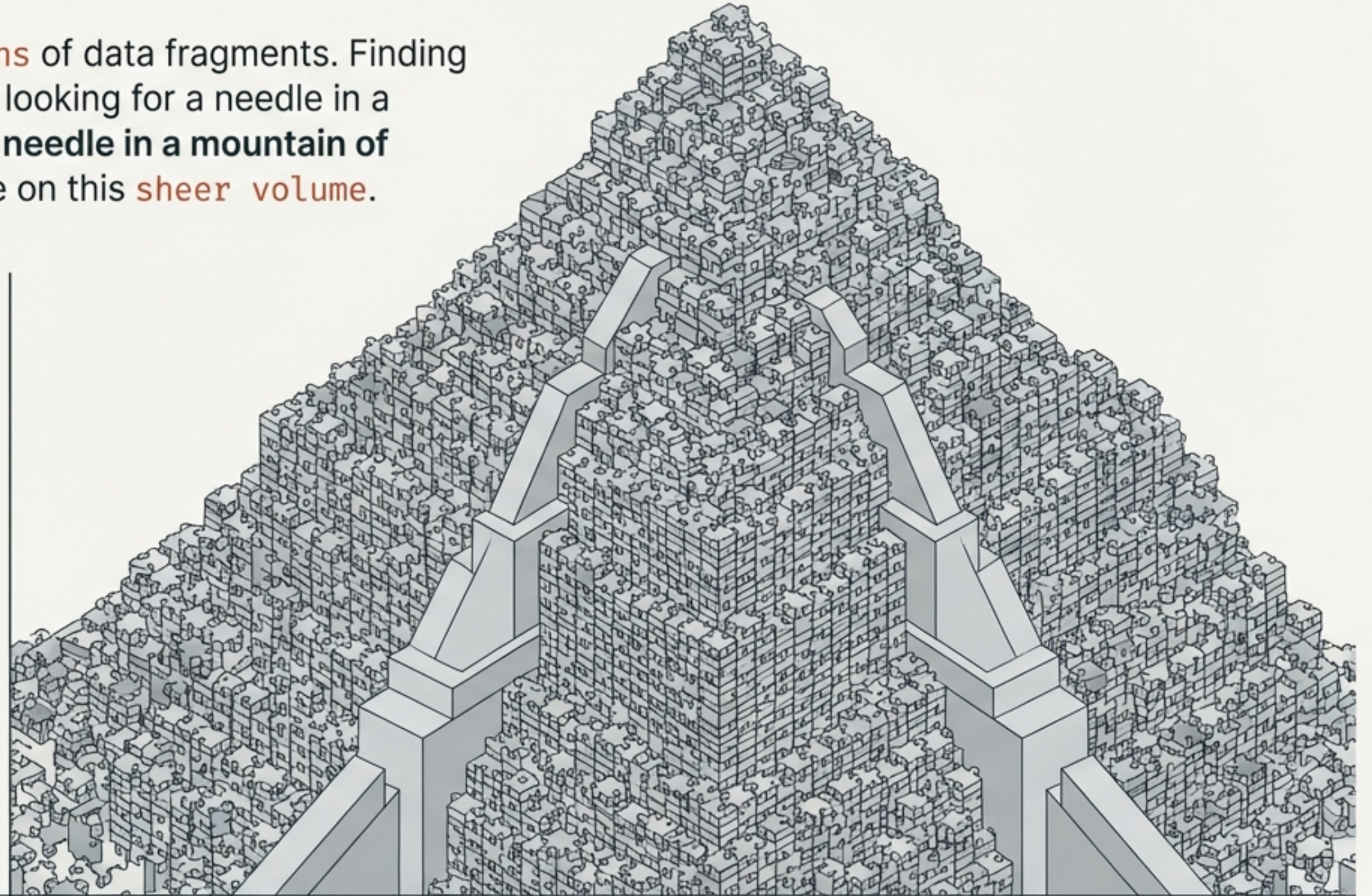


The Genomic Data Avalanche

Modern sequencing creates **billions** of data fragments. Finding a specific genetic mutation isn't just looking for a needle in a haystack—it's **looking for a specific needle in a mountain of other needles**. Standard tools choke on this **sheer volume**.



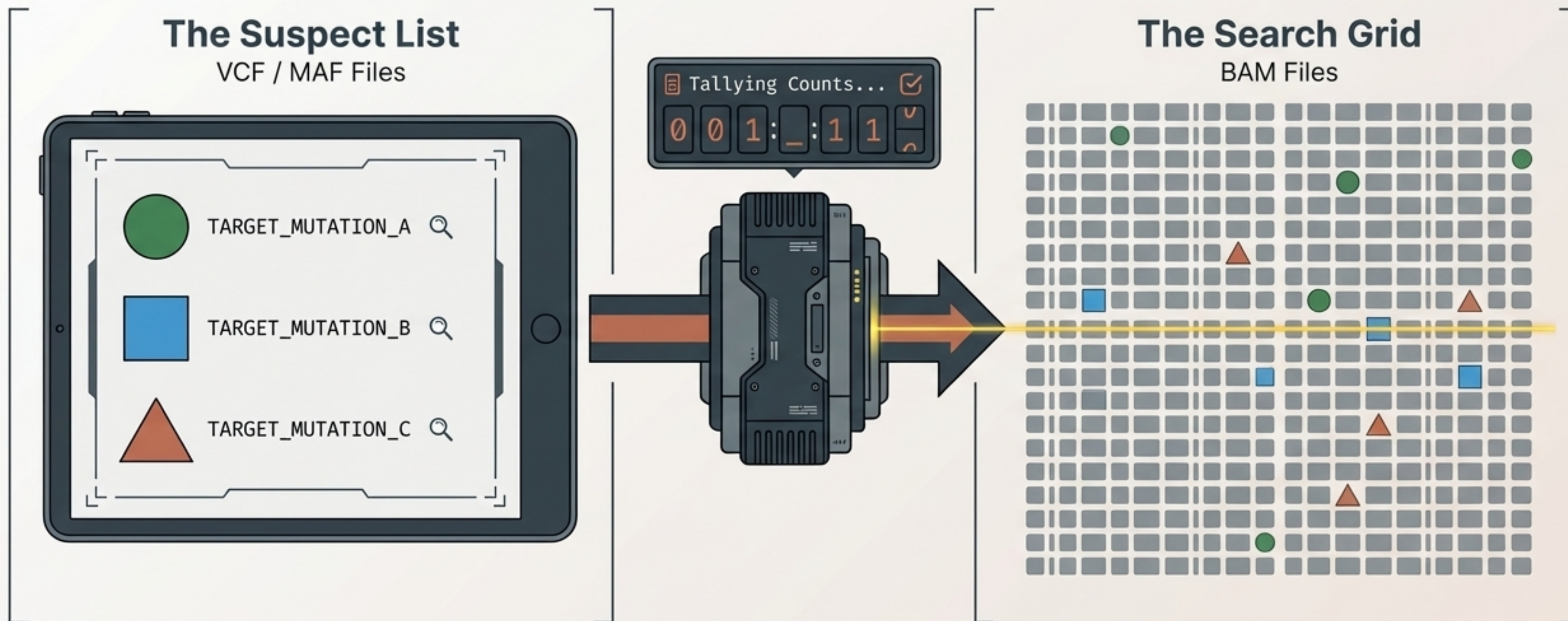
One Target Mutation



Billions of Raw Data Reads

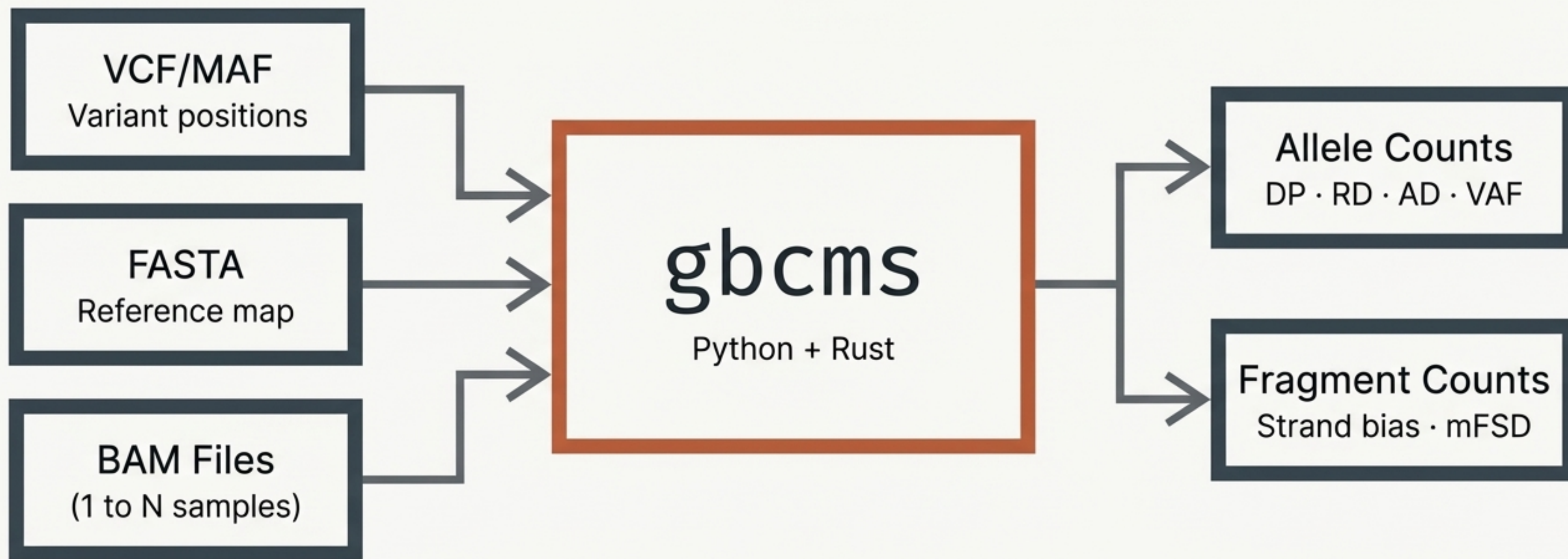
The Objective: Variant Counting

We have a list of **known** suspects (mutations). We need to scan the massive data pile to see **exactly if and how many times** they appear, generating a reliable frequency score.



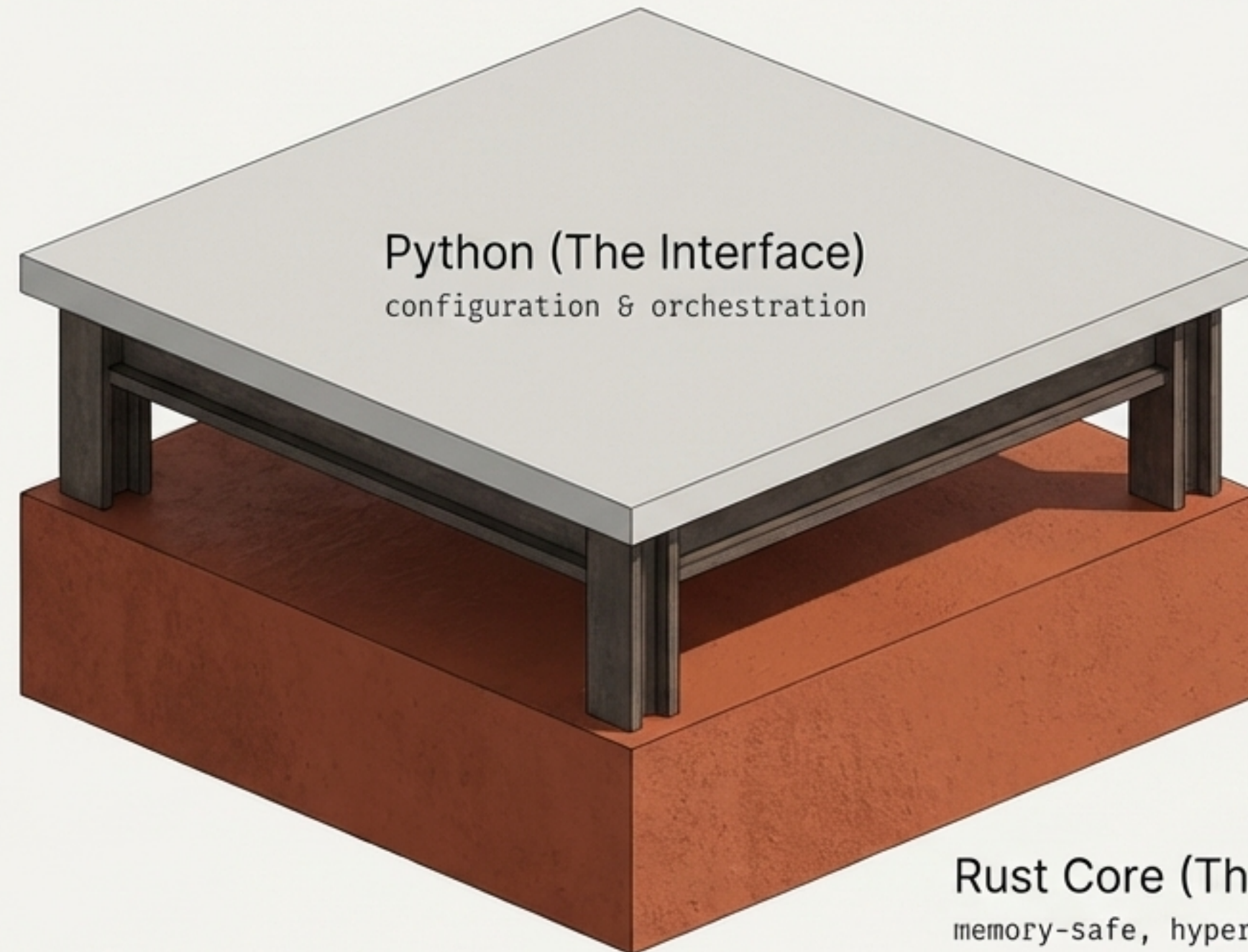
The Automated Sorting Factory

gbcms acts as an ultra-fast, automated sorting facility. It ingests raw sequencing data and target lists, filters out the noise, and outputs refined, immediately usable clinical metrics.



The Bilingual Advantage

A hybrid architecture combining ease of use with maximum processing power.

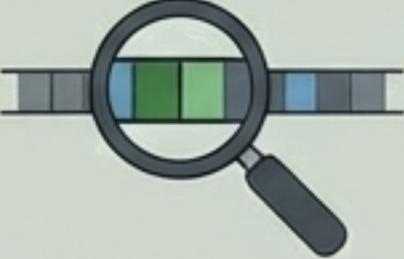
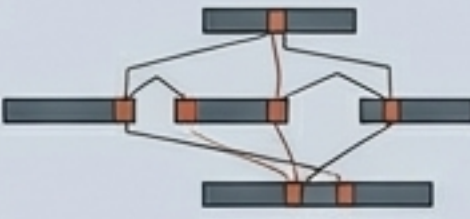


Makes the tool highly accessible, easy to command, and simple to integrate into existing workflows. Handles configuration and orchestration.

A memory-safe, hyper-optimized core that processes raw genomic data and mathematical algorithms at blistering speeds, yielding a **20x+ performance** improvement.

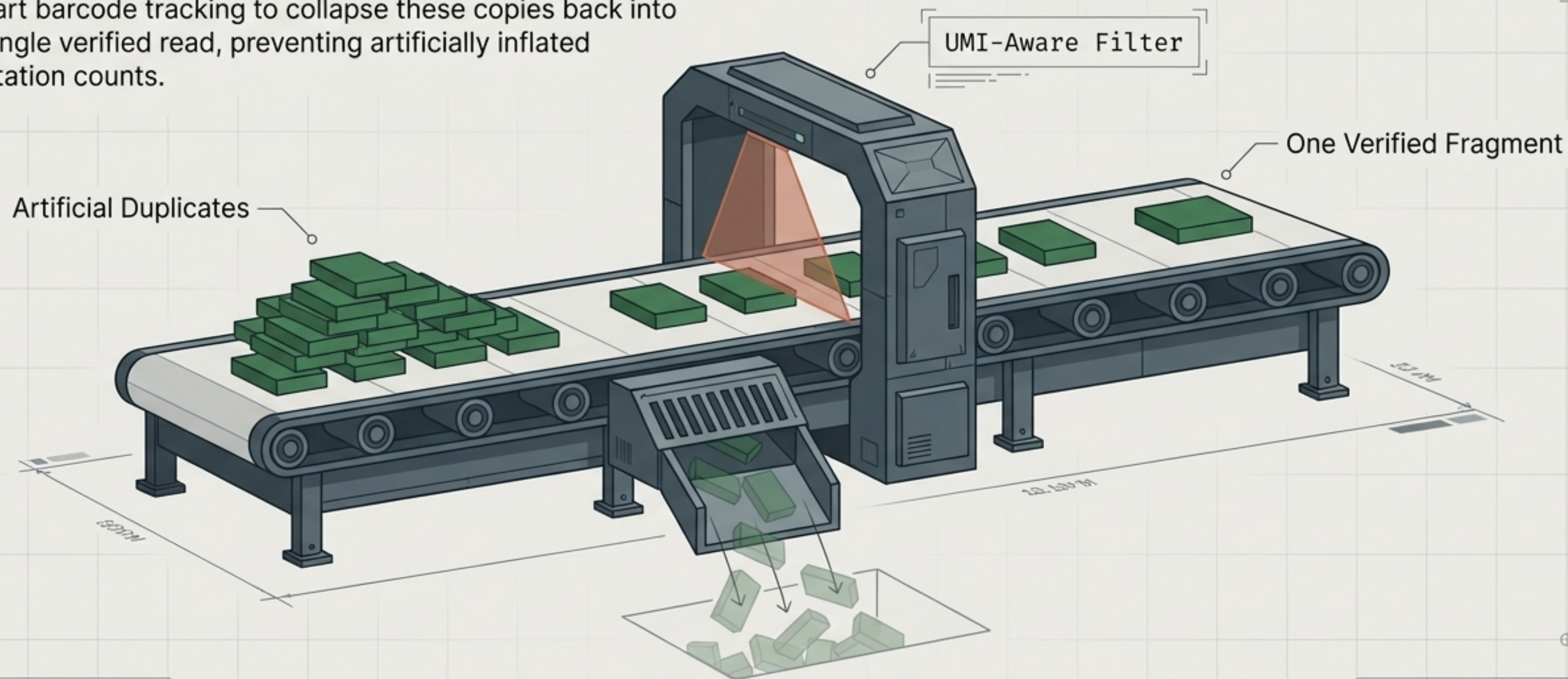
Two Distinct Assembly Lines

Counting mutations in static DNA requires a fundamentally different approach than tracking them in dynamic, spliced RNA.

DNA Mode	RNA Mode
 Checking the static original blueprint.	 Checking the dynamic, edited output.
 Navigating through chemical duplication artifacts and identical copies generated during sequencing.	 Navigating genes that have been cut, pasted, and naturally edited by the body over time.
 Strict deduplication, windowed indel detection, and exact spatial alignment to prevent false counts.	 Transcriptome-aware tracking, splice-junction mapping, and automatic flagging of natural A-to-I biological edits.

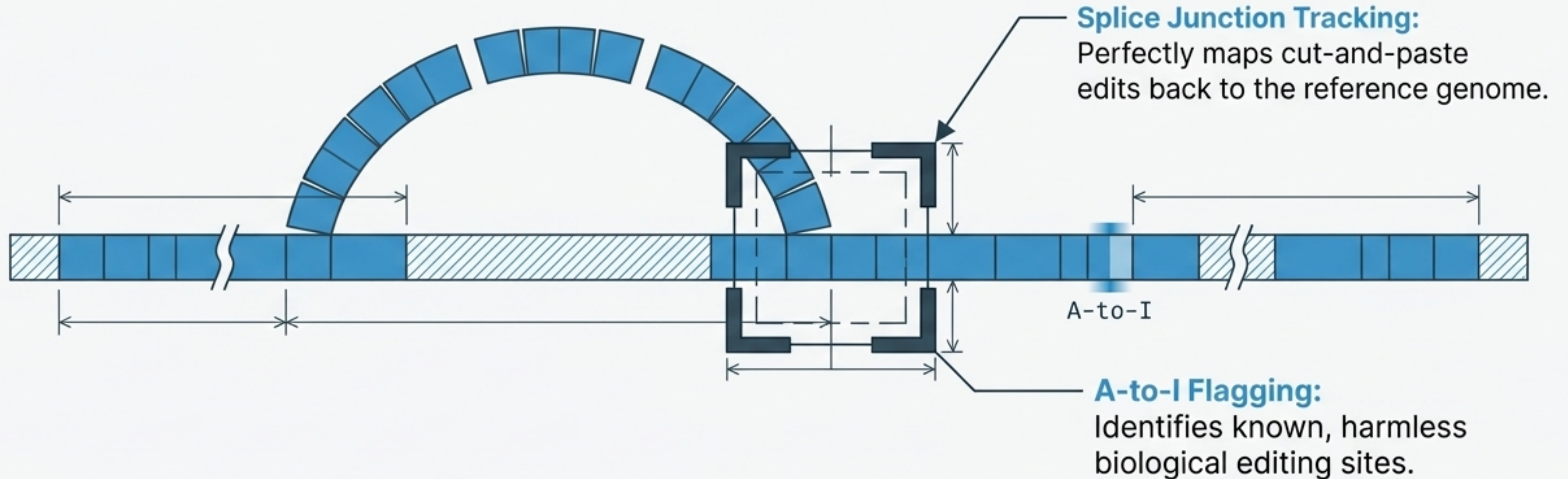
DNA Mode: Finding the True Signal

Sequencing often creates artificial duplicates—like making thousands of photocopies of a single page. gbcms uses smart barcode tracking to collapse these copies back into a single verified read, preventing artificially inflated mutation counts.



RNA Mode: Tracking the Edits

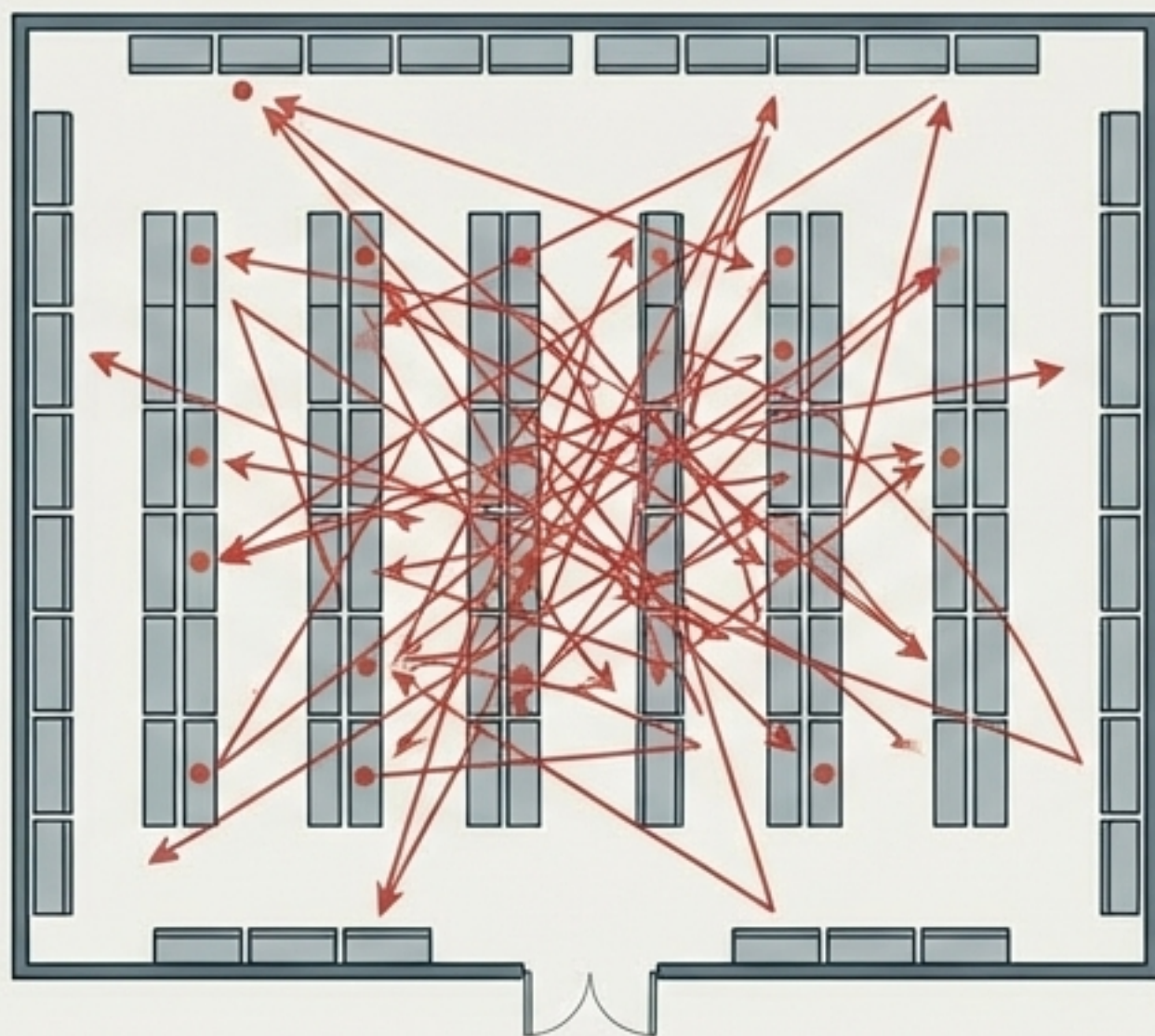
The body naturally edits RNA by cutting and pasting segments together. gbcms features dedicated tracking to map these natural cuts, ensuring normal biological processes aren't mistaken for dangerous disease mutations.



Smart Processing: Genomic Binning

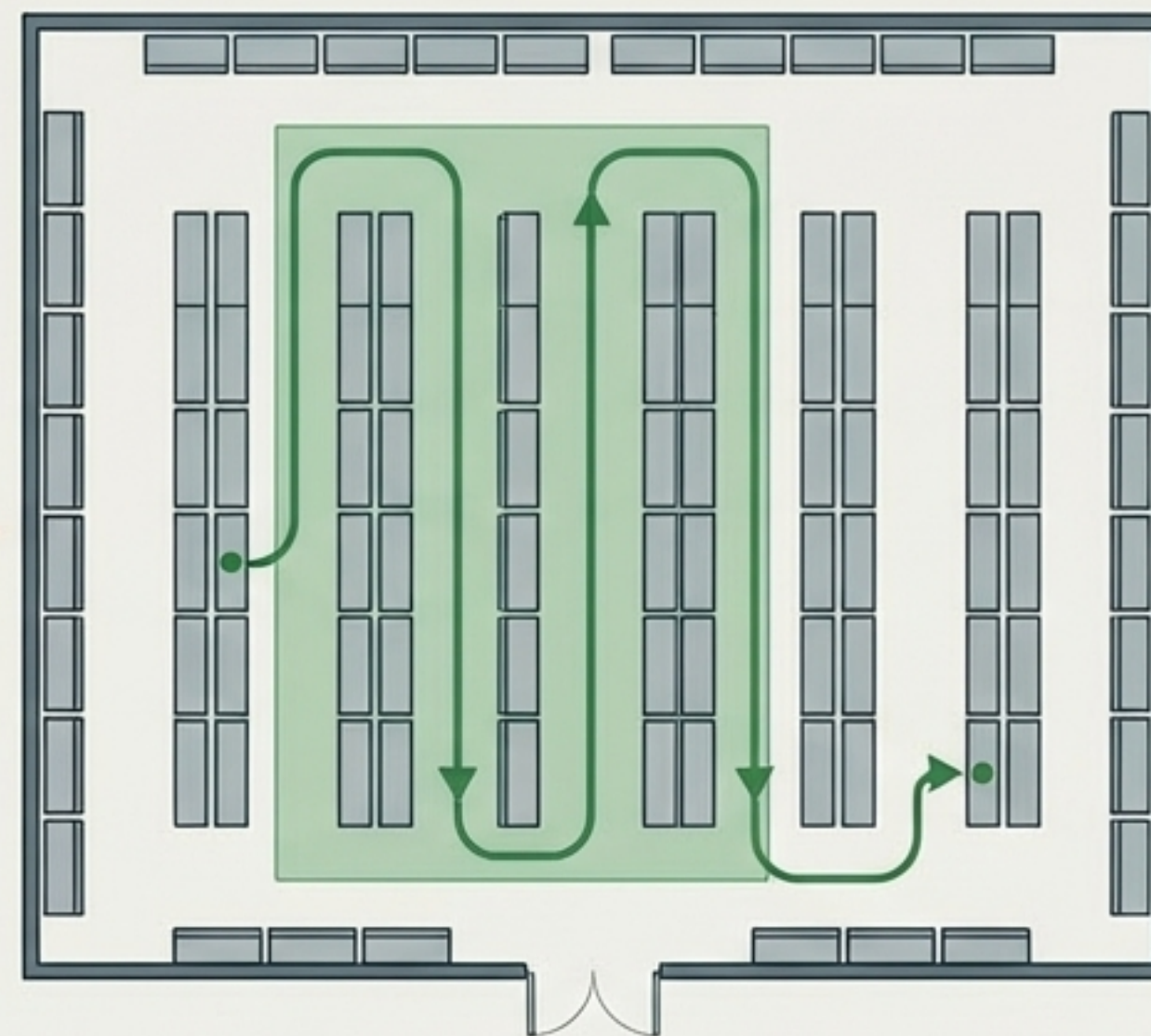
Instead of searching the massive data file from scratch for every single mutation (like running across a grocery store for each item), **gbcms** groups nearby targets together and resolves them simultaneously.

Standard Traversal (Inefficient)



Reduces disk
reads and I/O
operations by
5x to 20x.

~10kb Genomic Bins



Ironclad Accuracy & Error Correction

gbcms uses a 3-layer safeguard system to automatically exclude sequencing noise and mathematically establish the absolute truth of a mutation.

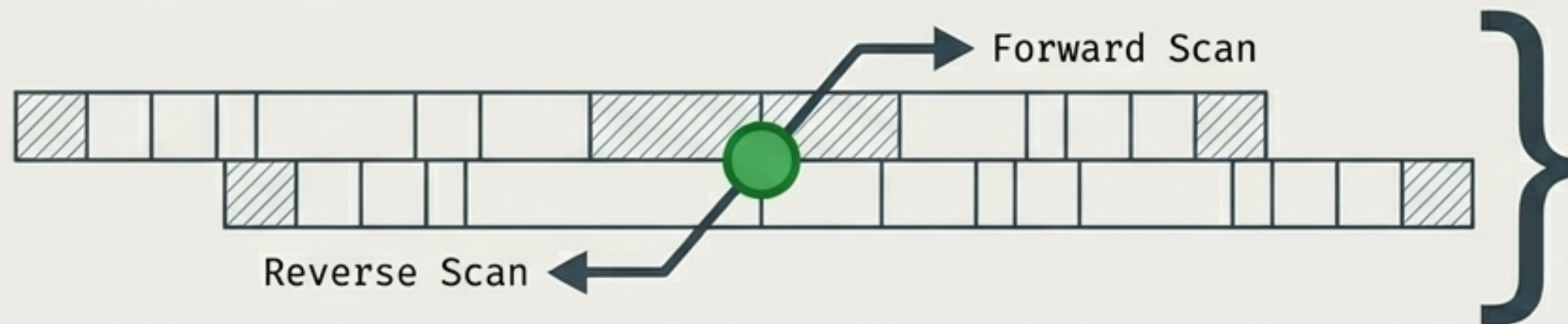
Masked Comparison



Masked Comparison:

Unreliable data is mathematically excluded so it cannot trigger as positive.

Fragment Consensus



Fragment Consensus:

Overlapping forward and reverse scans must agree to establish a verified count.

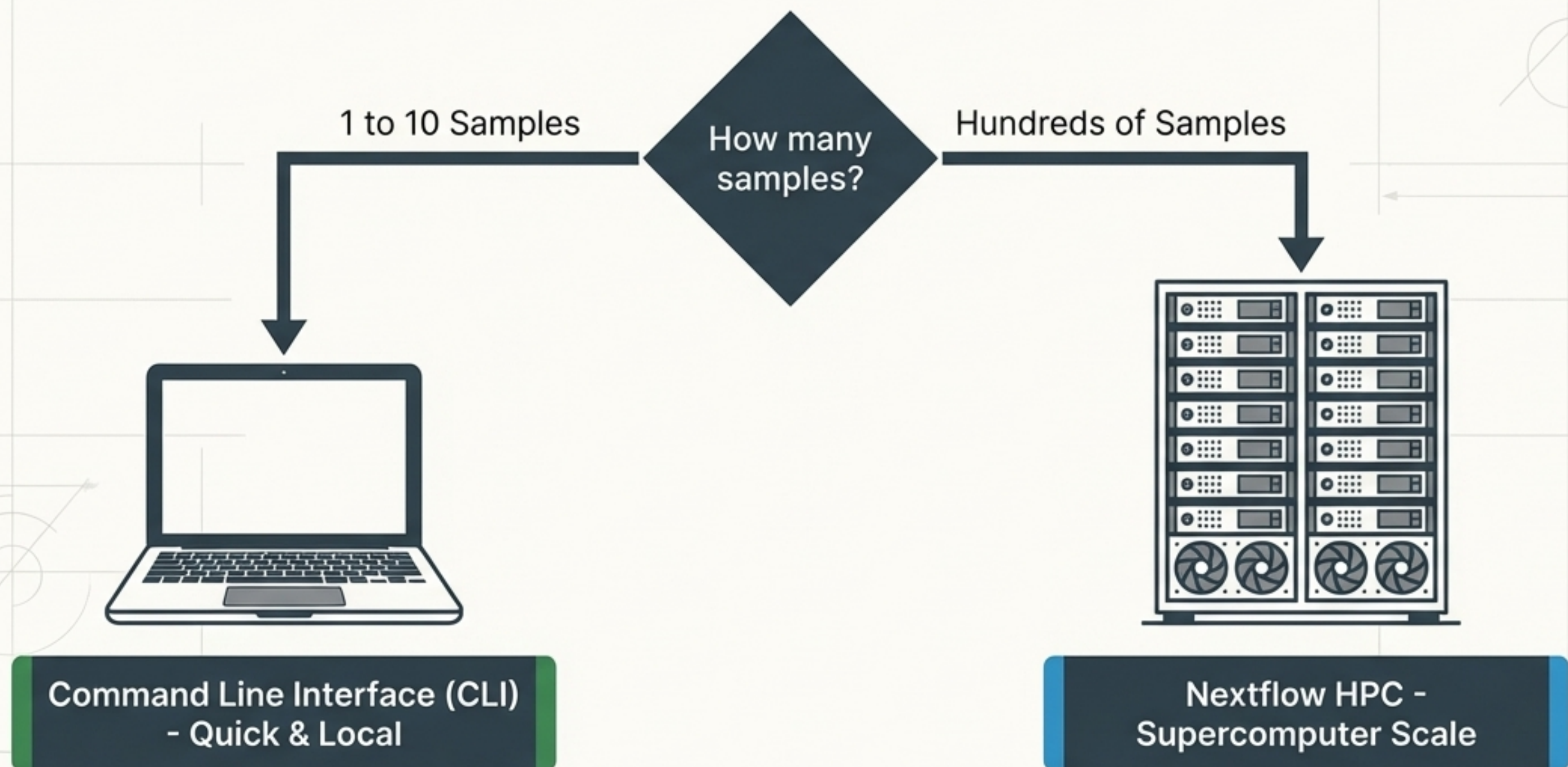
The Generational Leap

A complete rewrite engineered for unmatched performance, rigorous correctness, and modern bioinformatics workflows.

Feature	Original Software	The New gbcms
Target Scanning	Exact position match only	Smart Windowed Scan ($\pm 5\text{bp}$): Catches shifted mutations
Quality Control	No base quality threshold	Masked Comparison: Mathematically excludes noisy data
RNA Support	Not available	Dedicated transcriptome-aware filtering
Speed Engine	Basic blocked computing	Hybrid Python/ Rust engine via Rayon parallel processing

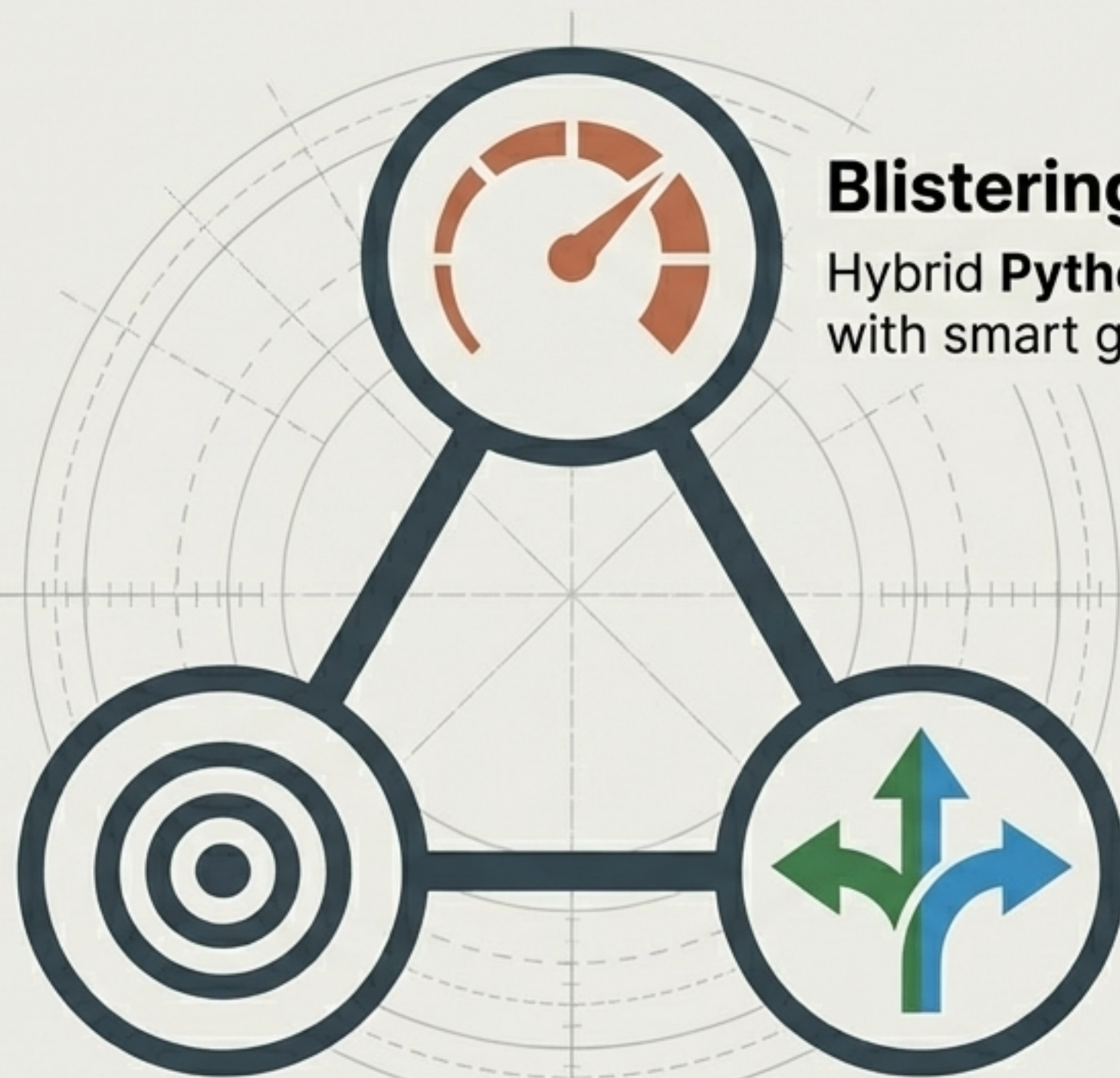
From Laptops to Supercomputers

Whether running a quick diagnostic on a local machine or processing thousands of samples simultaneously, the architecture scales to fit your infrastructure.



The Modern Standard for Variant Counting

Rigorous Accuracy
3-layer safeguards and
UMI-aware error correction.



Blistering Speed

Hybrid **Python/Rust** engine
with smart genomic binning.

Ultimate Flexibility

Dedicated DNA and RNA
assembly lines, scaling from
laptops to HPC clusters.